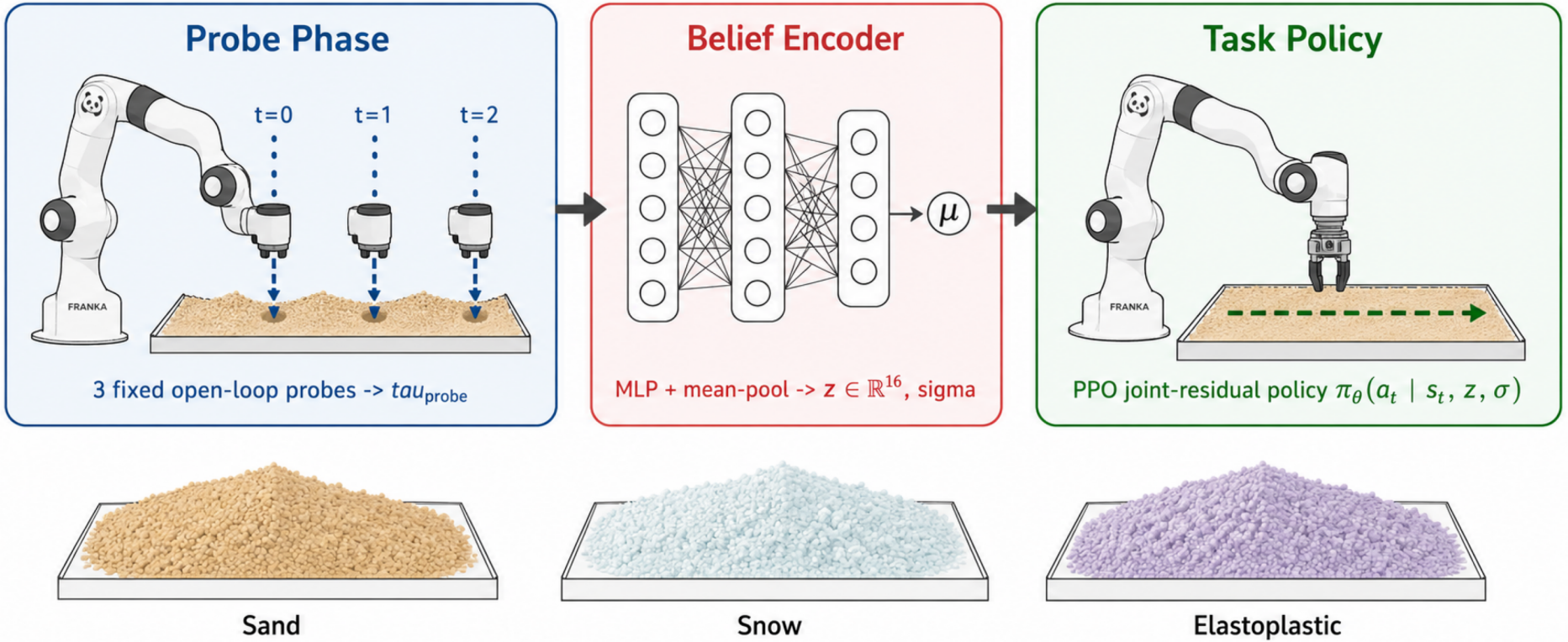
Probe Phase
t=0
t=1
t=2
FRANKA
3 fixed open-loop probes -> tau_{probe}
Belief Encoder
μ
MLP + mean-pool -> $z \in \mathbb{R}^{16}$, sigma
Task Policy
FRANKA
PPO joint-residual policy $\pi_\theta(a_t \mid s_t, z, \sigma)$
Sand
Snow
Elastoplastic
Hidden material properties: E, nu, rho, cohesion